

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Debugging and optimization techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Debugging & Optimization Task

Code of Conduct:

I Rishikesh K.S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K. Rishikesh

77/100

SIMATS ENGINEERING

ASSESSMENT TOOL 3

DEBUGGING & OPTIMISATION TASK

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Course	CSA07 – Computer Networks

QUESTION 1

A network administrator has configured a workstation with the following settings:

IP Address: 192.168.10.65

Subnet Mask: 255.255.255.192 (/26)

Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task — Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

1. Given Data

Parameter	Given Value
IP Address	192.168.10.65
Subnet Mask	255.255.255.192 (/26)
Default Gateway	192.168.10.1
Intended Subnet	192.168.10.0/26

PROBLEM IDENTIFICATION — WHAT IS WRONG AND WHY?

- 192.168.10.65 does not belong to the stated subnet 192.168.10.0/26.
- For /26, the block size is 64, so the subnet containing .65 starts at .64.
- The gateway 192.168.10.1 belongs to the .0/26 subnet, not the actual .64/26 subnet.
- Therefore the workstation IP and gateway are in different subnets.

2. Step-by-Step Debugging

Step 1 — Calculate the block size

$$256 - 192 = 64$$

Step 2 — List the /26 subnet ranges

$$192.168.10.0-.63 \mid .64-.127 \mid .128-.191 \mid .192-.255$$

Step 3 — Locate the configured IP

$$.65 \text{ lies in } .64-.127$$

$$\text{Network} = 192.168.10.64/26$$

$$\text{Valid hosts} = 192.168.10.65-192.168.10.126$$

$$\text{Broadcast} = 192.168.10.127$$

Step 4 — Check the gateway

$$192.168.10.1 \in 192.168.10.0/26$$

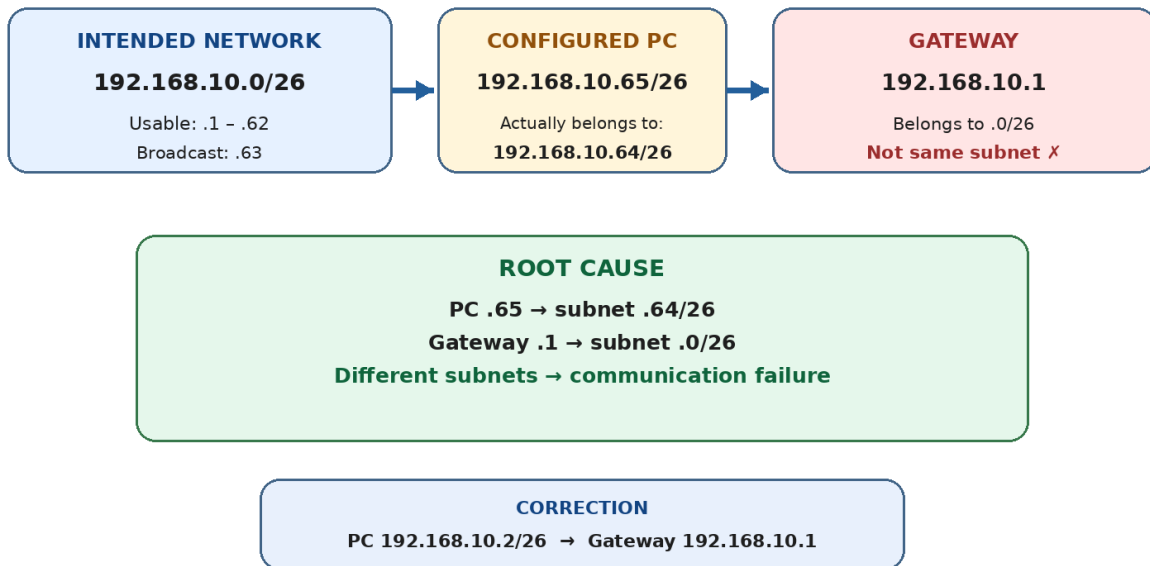
$$192.168.10.1 \notin 192.168.10.64/26$$

Step 5 — Root cause

The workstation is configured with an address from the 192.168.10.64/26 subnet but uses a gateway from the 192.168.10.0/26 subnet.

IPv4 Configuration → Find the Mismatch

Subnet: 192.168.10.0/26



3. Optimization / Correct Configuration

If the workstation must remain in the intended 192.168.10.0/26 subnet, assign a valid host from .1 to .62 and keep the gateway at .1.

Example: PC = 192.168.10.2/26

Gateway = 192.168.10.1

4. Usable Host Calculation

Host bits = $32 - 26 = 6$

Total addresses = $2^6 = 64$

Usable hosts = $64 - 2 = 62$

5. Final Answer

Item	Correct Result
Actual subnet of .65	192.168.10.64/26
Network address	192.168.10.64
Valid host range	.65–.126
Broadcast	.127
Gateway .1	Wrong for actual .64/26 subnet
Correct example	192.168.10.2/26 + gateway .1
Usable hosts	62

Analysis: The error is not guessed; it is proved by calculating the subnet boundary, checking the gateway, and then selecting a valid corrected configuration.

QUESTION 2

Two devices are configured with the following IPv6 addresses:

Device A: 2001:db8::ff00:42:8329/64

Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task — Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

1. Given Data

Device	IPv6 Address	Prefix
A	2001:db8::ff00:42:8329	/64
B	2001:db8:0:0:1::1	/64

PROBLEM IDENTIFICATION — WHAT IS WRONG AND WHY?

- Device A and Device B have different /64 network prefixes.
- The first 64 bits identify the subnet, so the prefix must be compared after expansion.
- Therefore, the devices are not in the same /64 subnet even though they are physically connected to the same network.

2. Step-by-Step Debugging

Step 1 — Expand Device A

2001:db8::ff00:42:8329 = 2001:0db8:0000:0000:0000:ff00:0042:8329

Step 2 — Expand Device B

2001:db8:0:0:1::1 = 2001:0db8:0000:0000:0001:0000:0000:0001

Step 3 — Determine /64 prefixes

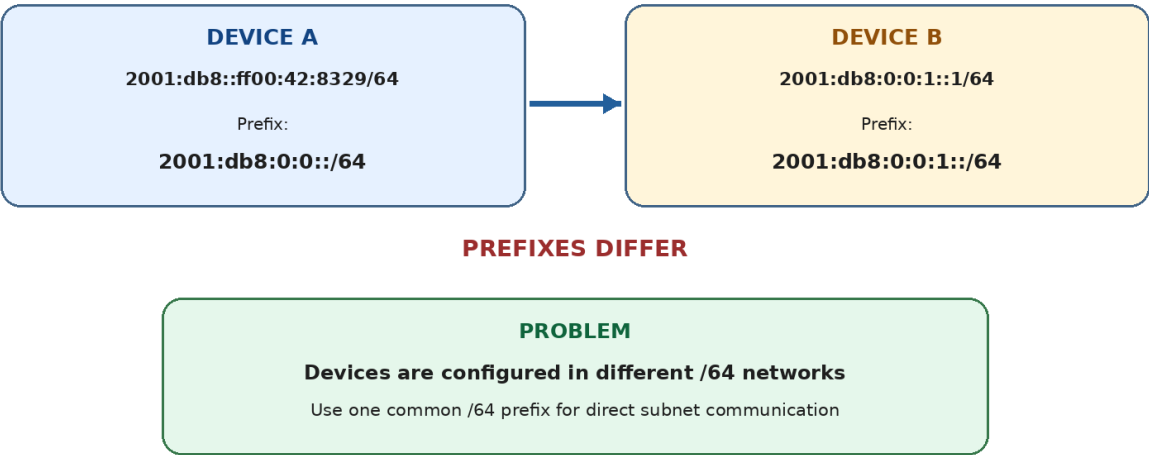
A = 2001:db8:0:0::/64

B = 2001:db8:0:0:1::/64

Step 4 — Compare

A prefix ≠ B prefix

IPv6 Prefix Check



3. Optimization

Common subnet = 2001:db8:0:0::/64
A = 2001:db8:0:0::10/64
B = 2001:db8:0:0::20/64

4. Host Address Calculation

Host bits = 128 - 64 = 64
Addresses = 2^64 = 18,446,744,073,709,551,616

5. Final Answer

Finding	Result
A prefix	2001:db8:0:0::/64
B prefix	2001:db8:0:0:1::/64
Same subnet?	No
Correction	Use common /64
Address space	2^64

Analysis: The mismatch is demonstrated by expanding the addresses and comparing the first four groups.

Q3. VLSM SUBNET ALLOCATION ANALYSIS

QUESTION

An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses. Task — Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

1. Given Data

Network	Masks Mentioned	Purpose
192.168.1.0/24	/26, /27, /28	Department subnetting
/26	64 total / 62 usable	Larger department
/27	32 total / 30 usable	Medium department
/28	16 total / 14 usable	Smaller department

PROBLEM IDENTIFICATION — WHAT IS WRONG AND WHY?

- The /24 network has enough addresses, but fixed subnet sizes may not match the real size of each department.
- A /26 gives 62 usable hosts, while /28 gives only 14 usable hosts. Giving a large block to a small department wastes addresses; giving a small block to a large department causes shortage.
- The question does not provide the actual host requirement of each department. Therefore, exact 'unused IP' counts and a unique final VLSM map cannot be calculated without inventing missing data.
- The correct approach is to calculate each block's capacity, identify the mismatch, then apply VLSM according to the actual departmental requirements.

2. Step-by-Step Calculation

Step 1 — /26 subnet

Network = 192.168.1.0/26

Range = .0–.63 | Hosts = .1–.62 | Broadcast = .63

Usable = 64 – 2 = 62

Step 2 — /27 subnet

Next aligned block = 192.168.1.64/27

Range = .64–.95 | Hosts = .65–.94 | Broadcast = .95

Usable = 32 – 2 = 30

Step 3 — /28 subnet

Next aligned block = 192.168.1.96/28

Range = .96–.111 | Hosts = .97–.110 | Broadcast = .111

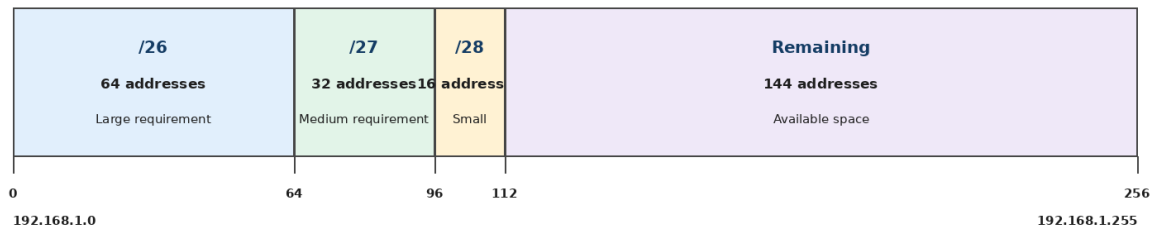
Usable = 16 – 2 = 14

Step 4 — Remaining address space

192.168.1.112–192.168.1.255 = 144 addresses remaining

VLSM — Efficient Address Allocation

Base Network: 192.168.1.0/24 | 256 total addresses



HOW THE BLOCK SIZES ARE DECIDED

$/26 \rightarrow 2^6 = 64 \text{ total} \rightarrow 62 \text{ usable}$ $/27 \rightarrow 2^5 = 32 \text{ total} \rightarrow 30 \text{ usable}$ $/28 \rightarrow 2^4 = 16 \text{ total} \rightarrow 14 \text{ usable}$

PROBLEM

Fixed-size allocation can waste addresses or cause shortages.

VLSM: allocate the largest requirement first, then smaller blocks.

3. Unused Address Analysis

The exact number of unused host addresses inside each department cannot be calculated from the supplied question because the actual number of hosts required/used by each department is not given.

Unused host addresses = allocated usable hosts - actual hosts used

Subnet	Usable Hosts	What can be concluded
/26	62	Up to 62 hosts can be supported
/27	30	Up to 30 hosts can be supported
/28	14	Up to 14 hosts can be supported

4. VLSM Optimization

Use the smallest subnet that satisfies the actual requirement, allocating the largest requirement first.

Actual Requirement	Smallest Suitable Prefix	Usable Hosts
1–14 hosts	/28	14
15–30 hosts	/27	30
31–62 hosts	/26	62
63–126 hosts	/25	126

Example only: if departments need 50, 25 and 10 hosts, use /26, /27 and /28 respectively. This is an illustration because those host counts are not provided in the original question.

5. Final Answer

The verified subnet calculations are $/26 \rightarrow 62$ usable, $/27 \rightarrow 30$ usable, and $/28 \rightarrow 14$ usable. The exact inefficient allocation and exact unused counts require the missing departmental host requirements. VLSM is the appropriate optimization method.

Analysis: The answer calculates every stated subnet, identifies the information gap instead of inventing data, and provides a correct VLSM decision rule.

Q4. IPv6 CONFIGURATION — DATA VERIFICATION

QUESTION

Two devices are configured with the following IPv6 addresses:

Device A: 2001:db8::ff00:42:8329/64

Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task — Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

1. Given Data

Device	Address	Actual /64 Prefix
A	2001:db8::ff00:42:8329	2001:db8:0:0::/64
B	2001:db8:0:0:1::1	2001:db8:0:0::/64

PROBLEM IDENTIFICATION — WHAT IS WRONG AND WHY?

- The supplied data does not show a /64 prefix mismatch.
- Both addresses share the first four groups: 2001:0db8:0000:0000.
- Therefore, based strictly on the given addresses, both devices belong to the same /64 subnet.
- The communication failure must be investigated using other checks such as interface status, VLAN, routing, duplicate-address detection, or ACL/firewall settings.

2. Step-by-Step Verification

Step 1 — Expand A

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 2 — Expand B

2001:0db8:0000:0000:0001:0000:0000:0001

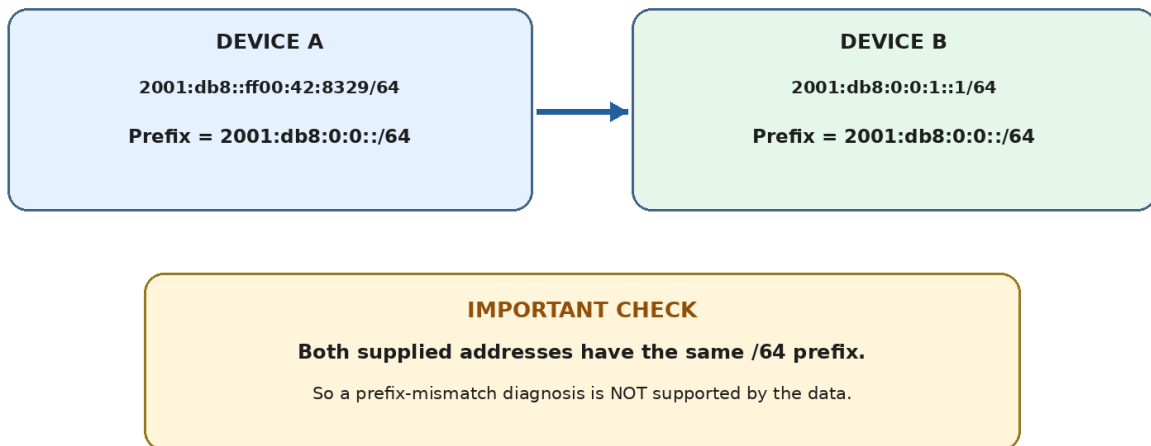
Step 3 — Compare /64 prefixes

A = 2001:db8:0:0::/64

B = 2001:db8:0:0::/64

Prefixes MATCH

IPv6 Prefix Verification



3. Further Debugging

- Check interface/link status.
- Check VLAN and Layer-2 connectivity.
- Check IPv6 routing entries.
- Check Duplicate Address Detection and address assignment.
- Check ACL/firewall rules.

4. Host Address Calculation

$$\text{Host bits} = 128 - 64 = 64$$

$$\text{Address space} = 2^{64}$$

5. Final Answer

Finding	Result
Prefix A	2001:db8:0:0::/64
Prefix B	2001:db8:0:0::/64
Prefix mismatch?	No
Next action	Debug other connectivity layers
Address space	2^{64}

Analysis: This answer preserves the source data and does not invent a prefix mismatch that the given addresses do not support.

QUESTION 5

A router contains the following routing table.

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task — Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

1. Given Data

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route 0.0.0.0/0	10.0.0.1

PROBLEM IDENTIFICATION — WHAT IS WRONG AND WHY?

- The destination 192.168.2.100 must be compared with every available route.
- It belongs to 192.168.2.0/24, so the specific /24 route should be selected.
- The default route is less specific and is used only when no more-specific route matches.

2. Step-by-Step Debugging

Step 1 — Check 192.168.1.0/24

192.168.2.100 → NO MATCH

Step 2 — Check 192.168.2.0/24

192.168.2.100 → MATCH

Step 3 — Longest-prefix rule

/24 is more specific than /0

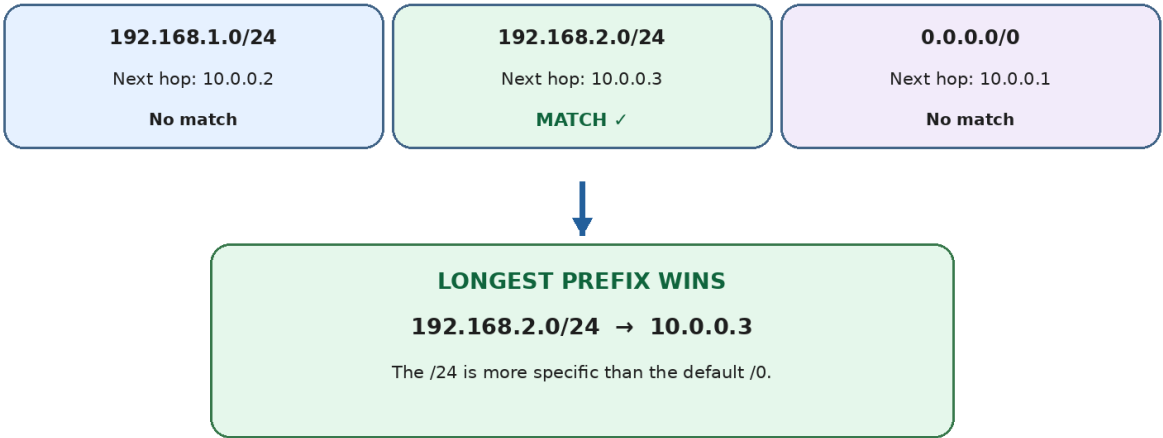
Selected next hop = 10.0.0.3

Step 4 — Default route

If no specific route matches → 0.0.0.0/0 → 10.0.0.1

Longest-Prefix Routing Decision

Destination: 192.168.2.100



3. Optimization / Further Debugging

The supplied routing table contains the correct specific route. If packets still fail, verify next-hop reachability, outgoing interface, return route, ARP and ACL/firewall configuration.

4. Final Answer

Item	Answer
Matching route	192.168.2.0/24
Next hop	10.0.0.3
Default route	0.0.0.0/0 via 10.0.0.1
Longest prefix	/24
Visible table error	No direct error shown

Analysis: The selected route is justified using longest-prefix matching, and further debugging is directed toward the next hop instead of changing a correct route.